Historias de Usuario – SMS Sensor Monitoring System

Proyecto: Sistema de Monitoreo de Sensores

This document presents the user stories for the sensor monitoring system. The system allows registering sensors, receiving readings, analyzing them using different strategies, and notifying critical events to subscribed modules.

HU-01 — Register sensor

Story
As a monitoring center operator, I want to register new sensors in the system, so that their readings can be received and analyzed later.

Acceptance Criteria

- Only sensors of type Temperature, Smoke, or Pressure are allowed.
- The system does not allow duplicate sensors to be registered.
- The system confirms the successful registration of the sensor.
- The sensor becomes available to receive readings.

Related Requirements

RF-01 — Register sensor.

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HU-02 — Enter Sensor Reading

Story

As a monitoring center operator, I want to enter the value of a sensor reading, so that the system can analyze it and determine whether it is a critical event.

Acceptance Criteria

- The sensor must be previously registered.
- The entered value must be a valid numeric value.
- The system creates a reading object with a timestamp.

- The reading is sent to the analysis module.
- All system observers are automatically notified.

Related Requirements

- RF-02 — Enter reading
- RF-03 — Analyze reading
- RF-05 — Notify observers

HU-03 — Change Analysis Strategy

Story

As a monitoring center supervisor, I want to change the system's analysis mode, so that detection thresholds can be adapted according to the operational context.

Acceptance Criteria

The system provides three analysis strategies:

- Strict
- Relaxed
- Adaptive
- Changing the strategy does not require restarting the system.
- The new strategy applies starting from the next entered reading.
- The system displays the active strategy in the menu.

Related Requirements

RF-04 — Change analysis mode.

HU-04 — View Alert History

Story

As a security auditor, I want to consult the complete history of readings and analyses, so that I can verify the system's behavior during critical events.

Acceptance Criteria

The history displays:

- date and time (timestamp)
- sensor type
- recorded value
- strategy used
- analysis result
- It includes both normal and critical readings.
- The history can be accessed at any time from the menu.

Related Requirements

RF-06 — View log history.

HU-05 — Receive Alarm for Critical Event

Story

As a field technician, I want to receive a visible alarm when a critical value is detected, so that I can act quickly in a risk situation.

Acceptance Criteria

- The alarm is triggered only when the analysis indicates a critical state.
- The alarm displays:
  - sensor type
  - detected value
  - sequential alert number
- The alarm module works independently from the other modules.
- The display and log continue working even if the alarm fails.

Related Requirements

RF-05 — Notify observers.